Systematic review

Please select one of the options below to edit your record. Either option will create a new version of the record - the existing version will remain unchanged.

A list of fields that can be edited in an update can be found here

1. * Review title.

Give the title of the review in English

Impacts of COVID-19 pandemic on preterm birth: a retrospective cohort study and meta-analysis

2. Original language title.

For reviews in languages other than English, give the title in the original language. This will be displayed with the English language title.

3. * Anticipated or actual start date.

Give the date the systematic review started or is expected to start.

30/03/2022

4. * Anticipated completion date. [1 change]

Give the date by which the review is expected to be completed.

20/05/2022

5. * Stage of review at time of this submission. [1 change]

This field uses answers to initial screening questions. It cannot be edited until after registration.

Tick the boxes to show which review tasks have been started and which have been completed.

Update this field each time any amendments are made to a published record.

The review has not yet started: No

Review stage	Started	Completed
Preliminary searches	Yes	Yes
Piloting of the study selection process	Yes	Yes
Formal screening of search results against eligibility criteria	Yes	Yes
Data extraction	Yes	Yes
Risk of bias (quality) assessment	Yes	Yes
Data analysis	Yes	Yes

Provide any other relevant information about the stage of the review here.

6. * Named contact.

The named contact is the guarantor for the accuracy of the information in the register record. This may be any member of the review team.

Xiaodie Yao

Email salutation (e.g. "Dr Smith" or "Joanne") for correspondence:
Miss Yao

7. * Named contact email.

Give the electronic email address of the named contact.

xiaodieyao@163.com

8. Named contact address

PLEASE NOTE this information will be published in the PROSPERO record so please do not enter private information, i.e. personal home address

Give the full institutional/organisational postal address for the named contact.

Nanjing Medical University, Jiangsu210003, Nanjing, China

9. Named contact phone number.

Give the telephone number for the named contact, including international dialling code.

15212060312

10. * Organisational affiliation of the review.

Full title of the organisational affiliations for this review and website address if available. This field may be completed as 'None' if the review is not affiliated to any organisation.

School of Public Health, Nanjing Medical University, Nanjing, China

Organisation web address:

11. * Review team members and their organisational affiliations. [1 change]

Give the personal details and the organisational affiliations of each member of the review team. Affiliation refers to groups or organisations to which review team members belong.
NOTE: email and country now MUST be entered for each person, unless you are amending a published record.

Miss Xiaodie Yao. Nanjing Medical University, Nanjing, China
Dr Juan Wen. Women’ s Hospital of Nanjing Medical University , Nanjing, China
Dr LiJun Zhu. Women’ s Hospital of Nanjing Medical University, Nanjing, China
Dr Jing Yin. Women’ s Hospital of Nanjing Medical University, Nanjing, China

12. * Funding sources/sponsors.

Details of the individuals, organizations, groups, companies or other legal entities who have funded or sponsored the review.

None

Grant number(s)

State the funder, grant or award number and the date of award

13. * Conflicts of interest.

List actual or perceived conflicts of interest (financial or academic).

None

14. Collaborators.

Give the name and affiliation of any individuals or organisations who are working on the review but who are not listed as review team members. **NOTE: email and country must be completed for each person, unless you are amending a published record.**

15. * Review question.

State the review question(s) clearly and precisely. It may be appropriate to break very broad questions down into a series of related more specific questions. Questions may be framed or refined using PI(E)COS or similar where relevant.

How does the lockdown of the COVID-19 pandemic impact preterm birth compared to the pre-pandemic period?

16. * Searches.

State the sources that will be searched (e.g. Medline). Give the search dates, and any restrictions (e.g. language or publication date). Do NOT enter the full search strategy (it may be provided as a link or attachment below.)

We will search MEDLINE and EMBASE databases up to March 2022 and no language restrictions. We also search reference lists to identify any other potentially relevant studies. Before the final analyses, we will further search the databases to find new studies.

17. URL to search strategy.

Upload a file with your search strategy, or an example of a search strategy for a specific database, (including the keywords) in pdf or word format. In doing so you are consenting to the file being made publicly accessible.

Or provide a URL or link to the strategy. Do NOT provide links to your search **results**.

Do not make this file publicly available until the review is complete

18. * Condition or domain being studied.

Give a short description of the disease, condition or healthcare domain being studied in your systematic review.

We aim to investigate the effects of the COVID-19 pandemic on preterm delivery. During the COVID-19 pandemic period , different countries imposed different lockdown measures. We will divide preterm delivery into PTB(<37 weeks of gestation), VPTB(<32 weeks of gestation), and EPTB(<32 weeks of gestation) for further detail analyses.

19. * Participants/population.

Specify the participants or populations being studied in the review. The preferred format includes details of both inclusion and exclusion criteria.

Pregnant women laboured during the lockdown period of the COVID-19 pandemic, without COVID-19 infection.

20. * Intervention(s), exposure(s).

Give full and clear descriptions or definitions of the interventions or the exposures to be reviewed. The preferred format includes details of both inclusion and exclusion criteria.

The COVID-19 pandemic period as exposure will be defined as from the date or the month of the implementation of emergency lockdown measures to after the lockdown was eased.

21. * Comparator(s)/control.

Where relevant, give details of the alternatives against which the intervention/exposure will be compared (e.g. another intervention or a non-exposed control group). The preferred format includes details of both inclusion and exclusion criteria.

Pre-pandemic period as control will be defined as before lockdown measure implementation with equivalent population.

22. * Types of study to be included.

Give details of the study designs (e.g. RCT) that are eligible for inclusion in the review. The preferred format includes both inclusion and exclusion criteria. If there are no restrictions on the types of study, this should be stated.

Prospective and retrospective cohort designs will be included.

23. Context.

Give summary details of the setting or other relevant characteristics, which help define the inclusion or exclusion criteria.

Studies in hospital or institute will depend on what is reported, without country setting to restrict.

24. * Main outcome(s).

Give the pre-specified main (most important) outcomes of the review, including details of how the outcome is defined and measured and when these measurement are made, if these are part of the review inclusion criteria.

Primary outcome: preterm birth. Secondary outcomes: very preterm birth (< 32 weeks of gestation) and extremely preterm birth (< 28 weeks of gestation). The characteristics of studies will be used to subgroup analyses for reliability of results.

Measures of effect
We will extract(calculate) odds ratios or relative risks with 95% confidence interval according to what were reported in the studies.

25. * Additional outcome(s).

List the pre-specified additional outcomes of the review, with a similar level of detail to that required for main outcomes. Where there are no additional outcomes please state 'None' or 'Not applicable' as appropriate to the review

None

Measures of effect

26. * Data extraction (selection and coding).

Describe how studies will be selected for inclusion. State what data will be extracted or obtained. State how this will be done and recorded.

Studies reporting on preterm outcomes before and during the pandemic will be included. Then, the effect size was provided or could be calculated and published in English. Two investigators independently reviewed all the articles, and disagreements were resolved by consensus with the third author. Data on authors, publication date, study design, sample size, study population, pandemic period definition, pre-pandemic period definition, effect size, and other related information will be extracted.

27. * Risk of bias (quality) assessment.

State which characteristics of the studies will be assessed and/or any formal risk of bias/quality assessment tools that will be used.

This review will include prospective and retrospective cohort studies , and therefore the Newcastle Ottawa scale will be used to assess risk of bias. The studies with a score of 0 to 3 were considered to have a high risk of bias, 4 to 6 had a moderate risk of bias, and 7 to 9 had a low risk of bias. This part will assessed by two investigators by consensus.

28. * Strategy for data synthesis.

Describe the methods you plan to use to synthesise data. This **must not be generic text** but should be **specific to your review** and describe how the proposed approach will be applied to your data.
If meta-analysis is planned, describe the models to be used, methods to explore statistical heterogeneity, and software package to be used.

For each outcome , we will extract or calculate the summary odds ratios and 95% confidence intervals. Statistical heterogeneity among studies was evaluated by using the $\chi 2$ test, I^2 statistics, and P values. If the homogeneous test was not significant ($I^2 > 50\%$), a fixed-effects model was used to obtain effect size. Otherwise, a random-effects model was used. If adjusted OR was not given in the studies, crude OR was used.

29. * Analysis of subgroups or subsets.

State any planned investigation of 'subgroups' . Be clear and specific about which type of study or participant will be included in each group or covariate investigated. State the planned analytic approach.

The characteristics of included studies will be extracted possible as further covariate investigated.

30. * Type and method of review.

Select the type of review, review method and health area from the lists below.

Type of review

Cost effectiveness	No
Diagnostic	No
Epidemiologic	Yes
Individual patient data (IPD) meta-analysis	No
Intervention	No
Living systematic review	No
Meta-analysis	Yes
Methodology	No
Narrative synthesis	No
Network meta-analysis	No
Pre-clinical	No
Prevention	No
Prognostic	No
Prospective meta-analysis (PMA)	No
Review of reviews	No
Service delivery	No
Synthesis of qualitative studies	No

Systematic review	Yes
Other	No

Health area of the review

Alcohol/substance misuse/abuse	No
Blood and immune system	No
Cancer	No
Cardiovascular	No
Care of the elderly	No
Child health	No
Complementary therapies	No
COVID-19	Yes

For COVID-19 registrations please tick all categories that apply. Doing so will enable your record to appear in area-specific searches

- Chinese medicine
- Diagnosis
- Epidemiological
- Genetics
- Health impacts
- Immunity
- Long COVID
- Mental health
- PPE
- Prognosis
- Public health intervention
- Rehabilitation
- Service delivery
- Transmission
- Treatments

Vaccines

Other

Crime and justice	No
Dental	No
Digestive system	No
Ear, nose and throat	No
Education	No
Endocrine and metabolic disorders	No
Eye disorders	No
General interest	No
Genetics	No
Health inequalities/health equity	No
Infections and infestations	No
International development	No
Mental health and behavioural conditions	No
Musculoskeletal	No
Neurological	No
Nursing	No
Obstetrics and gynaecology	No
Oral health	No
Palliative care	No
Perioperative care	No
Physiotherapy	No
Pregnancy and childbirth	Yes
Public health (including social determinants of health)	No
Rehabilitation	No
Respiratory disorders	No
Service delivery	No

Skin disorders	No
Social care	No
Surgery	No
Tropical Medicine	No
Urological	No
Wounds, injuries and accidents	No
Violence and abuse	No

31. Language.

Select each language individually to add it to the list below, use the bin icon to remove any added in error.

English

There is not an English language summary

32. * Country.

Select the country in which the review is being carried out. For multi-national collaborations select all the countries involved.

China

33. Other registration details.

Name any other organisation where the systematic review title or protocol is registered (e.g. Campbell, or The Joanna Briggs Institute) together with any unique identification number assigned by them.
If extracted data will be stored and made available through a repository such as the Systematic Review Data Repository (SRDR), details and a link should be included here. If none, leave blank.

34. Reference and/or URL for published protocol.

If the protocol for this review is published provide details (authors, title and journal details, preferably in Vancouver format)

No I do not make this file publicly available until the review is complete

35. Dissemination plans.

Do you intend to publish the review on completion?

No

36. Keywords.

Give words or phrases that best describe the review. Separate keywords with a semicolon or new line. Keywords help PROSPERO users find your review (keywords do not appear in the public record but are included in searches). Be as specific and precise as possible. Avoid acronyms and abbreviations unless these are in wide use.

preterm birth; COVID-19 pandemic; retrospective cohort

37. Details of any existing review of the same topic by the same authors.

If you are registering an update of an existing review give details of the earlier versions and include a full bibliographic reference, if available.

38. * Current review status. [1 change]

Update review status when the review is completed and when it is published.
New registrations must be ongoing so this field is not editable for initial submission.

Review_Completed_not_published

39. Any additional information.

Provide any other information relevant to the registration of this review.

40. Details of final report/publication(s) or preprints if available.

Leave empty until publication details are available OR you have a link to a preprint (NOTE: this field is not editable for initial submission).
List authors, title and journal details preferably in Vancouver format.